RailwayBooking.java

Producer-consumer style booking demo using wait()/notify()

Source Code

```
class Railway {
    int seats = 1;

    synchronized void book() {
        try {
            if (seats == 0) {
                System.out.println("No seats. Booking thread waiting...");
                wait();
            }
            seats--;
            System.out.println("Ticket booked.");
            System.out.println("Available seats: " + seats);
        } catch (Exception e) {}
    }

    synchronized void cancel() {
        seats++;
        System.out.println("Ticket cancelled.");
        System.out.println("Available seats: " + seats);
        notify();
    }
}

class Booking extends Thread {
    Railway r;
    Booking(Railway r) { this.r = r; }
    public void run() { r.book(); }
}

class Cancellation extends Thread {
    Railway r;
    Cancellation(Railway r) { this.r = r; }
    public void run() {
        try { Thread.sleep(1000); } catch (Exception e) {}
        r.cancel();
    }
}

public class RailwayBooking {
    public static void main(String[] args) throws Exception {
        Railway r = new Railway();

        Booking b1 = new Booking(r);
        b1.start();
        b1.join();

        Booking b2 = new Booking(r);
        b2.start();

        Cancellation c = new Cancellation(r);
        c.start();
    }
}
```

Output

```
Ticket booked.
Available seats: 0
No seats. Booking thread waiting...
Ticket cancelled.
Available seats: 1
Ticket booked.
Available seats: 0
```